
Practice Paper 3

Advanced Topics, Deep Learning & Modern GenAI

101 questions across 9 topics. Five options each (A-E). One correct answer per question. Answer key at the end.

Name: _____

Student No.: _____

Date: _____

Score: _____

Topic 1: Foundations & End-to-End ML Pipelines

Reading traps on bias-variance, no-free-lunch, CACE/entanglement, hidden feedback loops, and production systems

1 The lab09_rag corpus file 01_course_outline.md states DSA 8401 is assessed by continuous assessment (40%) plus a final exam (60%), with eight labs worth 5% each. The official syllabus handout instead lists Lab Assignments & Quizzes at 20%, a Midterm Exam at 20%, and a Capstone Project at 60%, with no traditional final exam. How should these two documents be reconciled?

- A. They are not reconciled as fact vs. fact — the RAG corpus is a separate synthetic knowledge base built for the retrieval lab exercise, while the syllabus handout is the authoritative record of this course's actual assessment structure; a retrieval system querying the corpus would report the corpus's own figures, not the real ones
- B. The syllabus handout is outdated and the RAG corpus reflects a later, corrected policy
- C. The two documents describe two entirely different, unrelated units
- D. The final exam was later replaced by the capstone, so both statements are simultaneously true at different points in the same semester
- E. The 40/60 split applies to undergraduates while the 20/20/60 split applies to graduate students in the same class

2 Per Section 1.3's account of the bias-variance trade-off, which pairing of (training error, test error, relationship between them) correctly characterises HIGH VARIANCE (overfitting), as distinct from high bias?

- A. Training and test error are both moderate and converge to the same value regardless of model capacity
- B. Training error is low; test error is high; a large gap separates them — the model is flexible enough to fit the noise of the particular training sample, and a different sample would have yielded a visibly different model
- C. Training error is high; test error is high; the two remain close together
- D. Training error is low; test error is also low, with the gap widening only after deployment
- E. Training error is high; test error is low, with the gap explained by a lucky test split

3 The course cautions that the no-free-lunch theorem 'is sometimes quoted nihilistically.' Which statement below IS an example of that nihilistic misreading, as opposed to the course's own practical reading of the theorem?

- A. Gradient-boosted trees often win on tabular data, but that is an empirical regularity re-verified on every project, not a theorem
- B. Benchmark leaderboard performance transfers to a new problem only insofar as the distributions share structure
- C. Because averaged over all possible problems every algorithm performs identically, there is no principled basis for preferring one model family over another on any given problem, so model selection is arbitrary
- D. Priors are unavoidable, so an engineer should choose them consciously based on the structure of the specific problem
- E. Because no family wins a priori, several diverse families should be shortlisted and compared via honest validation

4 A data scientist argues: 'Since we retrained only the fraud-score feature's binning logic and left every other feature untouched, we can assume the model's behaviour on every other input is unchanged.' Which principle most directly contradicts this reasoning, and why?

- A. Data leakage — because the binning logic must have used test-time information
- B. The bias-variance trade-off — because binning always reduces model bias with no effect on variance
- C. Batch vs. online learning — because binning changes are only meaningful in online systems
- D. CACE (entanglement) — because features interact through the model's learned weights, changing how one input is represented can silently change how the model uses every other feature, so 'other features untouched' does not imply 'other features' influence unchanged'
- E. The no-free-lunch theorem — because no feature is universally informative, the re-trained feature cannot have affected anything

5 A lending model declines low-score applicants. Six months later, an analyst reports: 'Our model's false-negative rate for declined applicants is only 2%, so the model rarely misses a good applicant.' What is the deepest flaw in this claim?

- A. The 2% figure is accurate, but only reflects accepted applicants, not declined ones, which is a minor and acceptable limitation
- B. The false-negative rate should instead be computed using the training set, not deployment data
- C. This is a data leakage problem, not a feedback loop problem
- D. The 2% figure is inflated because of the accuracy paradox on an imbalanced dataset
- E. The 2% figure cannot actually be computed honestly, because declined applicants' true repayment outcome is never observed — there is no ground truth available to measure a false-negative rate on that population at all

6 Per the course's systems diagram, which pairing correctly matches a system component to a plausible real failure mode in a deployed credit model?

- A.** Configuration — a silently misconfigured feature-store lookback window causes the RFM features computed in production to differ from those used during training, degrading performance without any code change
 - B.** ML code — a change to the serving infrastructure's load balancer alters the model's predicted probabilities
 - C.** Feature extraction — a GPU driver update in the serving cluster causes the model's coefficients to change
 - D.** Monitoring — an increase in monthly loan applications directly rewrites the trained model's weights
 - E.** Data verification — the model's learned weights are used to detect anomalies in the load balancer's configuration file
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7 Per the course's batch-vs-online comparison table, which statement correctly matches the 'attack surface' row?

- A.** Attack surface refers exclusively to adversarial examples crafted at inference time, not to training-time poisoning
 - B.** In batch learning, any poisoning of the training data must survive a retrain-and-review cycle before it affects the live model; in online learning, a poisoned stream can corrupt the live model directly and immediately
 - C.** In batch learning, a poisoned stream corrupts the live model directly; in online learning, poisoning must survive a retrain-and-review cycle
 - D.** Neither batch nor online learning is vulnerable to data poisoning, since both use validated pipelines
 - E.** Attack surface only applies to online learning; batch systems have no attack surface at all
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8 Which of the following correctly orders three stages of the ML lifecycle as taught in this course: (i) fine-tune and select, (ii) collect and prepare data, (iii) shortlist candidate model families?

- A.** All three happen simultaneously in a single step with no meaningful order
 - B.** (ii) collect and prepare data → (i) fine-tune and select → (iii) shortlist candidate model families
 - C.** (ii) collect and prepare data → (iii) shortlist candidate model families → (i) fine-tune and select
 - D.** (iii) shortlist candidate model families → (i) fine-tune and select → (ii) collect and prepare data
 - E.** (i) fine-tune and select → (ii) collect and prepare data → (iii) shortlist candidate model families
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- 9** In logistic regression, $C = 1/\lambda$. A practitioner increases C from 1.0 to 100.0. Which is the most accurate consequence?
- A. C has no relationship to λ ; they are independent hyperparameters
 - B. Increasing C always strictly improves test performance, since more flexibility is always better
 - C. Increasing C always strictly worsens test performance, since C directly measures overfitting
 - D. Regularization strength λ decreases as C increases, so the model is allowed more flexibility to fit the training data — training loss will tend to fall, but the effect on expected (test) risk depends on where the model currently sits on the bias-variance curve, and could go either way
 - E. Regularization strength λ increases as C increases, guaranteeing lower test error
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- 10** Per the schematic in the course material, which of these is depicted as PART of the small green 'ML code' core, rather than one of the surrounding system components?
- A. Feature extraction
 - B. Monitoring
 - C. Serving infrastructure
 - D. Resource management
 - E. None of the listed items — configuration, monitoring, data collection, serving infrastructure, feature extraction, data verification, resource management, and analysis tools are ALL depicted as surrounding the small green core, not part of it
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- 11** Per the batch-vs-online decision table, which pairing correctly matches a 'natural fit' example to its learning paradigm?
- A. Batch learning naturally fits credit scoring and limit setting; online learning naturally fits streaming fraud detection on payment rails
 - B. Batch learning naturally fits streaming fraud detection; online learning naturally fits credit scoring and limit setting
 - C. Both credit scoring and streaming fraud detection are described as natural fits exclusively for online learning
 - D. Both credit scoring and streaming fraud detection are described as natural fits exclusively for batch learning
 - E. The table does not name any natural-fit examples for either paradigm
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- 12** A team observes that after silently changing how a single categorical feature is encoded (no other change), overall model AUC stays flat, but the ranking of individual applicants shifts meaningfully within score bands, and a downstream fairness audit that was passing now flags a disparity. Which single foundational principle best explains why an apparently 'neutral' encoding change could ripple this far?
- A. Batch vs. online learning, since fairness audits only apply to online systems
 - B. CACE / entanglement — because features interact through the model's learned weights, a change to one feature's representation can silently redistribute the model's reliance across all correlated features, changing individual rankings and fairness properties even when an aggregate metric like AUC is unchanged
 - C. The no-free-lunch theorem, since no encoding is universally fair
 - D. Data leakage, since encoding changes always leak information from the test set
 - E. The bias-variance trade-off, since encoding changes only affect variance, not fairness
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Topic 2: Advanced Data Preparation & Feature Engineering

MCAR/MAR/MNAR subtleties, leakage-safe discipline, target encoding, entity resolution, cyclical encoding

- 13** device_id is classified as MCAR, agent_id as MAR, and gps_lat/gps_lon as MNAR. Which statement correctly explains why agent_id is MAR rather than MCAR?
- A. agent_id is MAR because it has more missing values than device_id
 - B. agent_id is MAR because it was one of the two columns dropped for leakage
 - C. agent_id's missingness can be explained by other OBSERVED variables (e.g., transaction channel or agent network coverage), whereas MCAR missingness must be unrelated to any variable, observed or not
 - D. agent_id is MAR because its missingness is caused directly by its own unobserved value, which is the defining property of MAR
 - E. agent_id is MAR because it is a string column, and all string columns are MAR by convention
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- 14** The course's formal definition of leakage involves a training-time feature map ϕ_{train} and a serving-time feature map ϕ_{serve} . Leakage exists precisely when:
- A. ϕ_{train} produces more columns than ϕ_{serve}
 - B. ϕ_{serve} is fit on the test set instead of the training set
 - C. ϕ_{train} and ϕ_{serve} disagree only in their column ordering, not their content
 - D. ϕ_{train} encodes information that would not be available to ϕ_{serve} at the moment of prediction — the two feature maps disagree in a way that favours training
 - E. ϕ_{train} and ϕ_{serve} are computed using different programming languages
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15 Target encoding is described as a shrinkage estimate of a Beta-Binomial model with prior strength m centred at the global target mean $\bar{y}$. What does increasing m do to a rare category's encoded value?

- A. It pulls the rare category's encoded value more strongly toward its own raw target mean, ignoring the global mean entirely
 - B. It has no effect on rare categories, only on frequent ones
 - C. It converts the encoding from a numeric value into a one-hot vector
 - D. It removes the need for out-of-fold encoding entirely, regardless of m
 - E. It pulls the rare category's encoded value more strongly toward the global mean $\bar{y}$, reducing the influence of that category's own (noisy, low-sample) target statistic
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16 Suppose a team resolves entities correctly using `reg_id`, but then evaluates using plain KFold instead of GroupKFold. What specific problem remains despite the correct entity resolution?

- A. Rows belonging to the same resolved customer can still be split across the training and validation folds, letting the model implicitly 'recognise' a customer's own patterns from training when evaluated on their held-out rows
 - B. Entity resolution and cross-validation are unrelated, so no problem remains once entities are resolved
 - C. Plain KFold would fail to run entirely on a dataset with a resolved `customer_id` column
 - D. The problem shifts from leakage to class imbalance
 - E. Entity resolution alone is sufficient to prevent leakage, since it deduplicates the customer identifiers
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17 A day-of-week feature is cyclically encoded as $(\sin(2\pi \cdot d/7), \cos(2\pi \cdot d/7))$ for $d = 0..6$. Why is a period of 7 used here rather than 24 (as used for hour-of-day)?

- A. The period is arbitrary and does not need to match the variable's actual cycle length
 - B. The cycle length must match the natural period of the variable being encoded — day-of-week repeats every 7 values, while hour-of-day repeats every 24, so each feature uses its own true period
 - C. 7 is always used regardless of the variable, since it is the standard default in scikit-learn
 - D. 24 must be used for both, since sine/cosine encoding requires the same period across all cyclical features in a pipeline
 - E. 7 is used because there are 7 principal components retained in this dataset
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18 The train/serving split discipline ('split, apply everywhere') is named as 'the single most effective leakage vaccine.' Which of the following would violate this discipline even though it might look harmless?

- A. Dropping `manual_review_score` and `settlement_status` before training
- B. Encoding categorical variables with one-hot encoding fit on the training split only
- C. Computing a customer's overall average transaction amount using their entire history — including transactions after the scoring timestamp — and using it as a feature for every row, including rows scored before those later transactions occurred
- D. Fitting a `StandardScaler` on the training split only, then applying it unchanged to the test split
- E. Using `GroupKFold` grouped by `customer_id` during cross-validation

19 In scorecard practice, each bin of each feature receives a 'weight of evidence' (WoE). What does a large POSITIVE WoE for a given bin typically indicate?

- A. That bin has more rows than any other bin in the feature
- B. That bin was excluded from the leakage audit
- C. That bin's raw numeric value (before binning) was negative
- D. That bin's proportion of 'good' (non-default) cases is much larger relative to 'bad' (default) cases than the overall population — the bin is associated with lower risk
- E. That bin contains missing values that could not be resolved

20 A feature contains transaction amounts that can be negative (debits) after sign-correction. A colleague suggests adding a constant offset to make every value positive and then applying Box-Cox, instead of switching to Yeo-Johnson. What is the strongest objection?

- A. Box-Cox cannot mathematically be applied to any dataset containing amounts, regardless of sign
- B. Adding a constant offset is mathematically identical to Yeo-Johnson, so there is no real objection
- C. Box-Cox only works on categorical features, not numeric ones
- D. The offset approach is fine and is exactly how Yeo-Johnson is implemented internally
- E. The offset is arbitrary and must be re-derived any time the data's minimum changes (e.g. with new data at scoring time), making the transform unstable and hard to keep leakage-safe across train/serve, whereas Yeo-Johnson handles zero/negative values natively

21 Two engineers compute a customer's 'frequency' RFM feature differently: Engineer A counts transactions up to and including SCORING_TS; Engineer B counts ALL transactions in the customer's history, including those after SCORING_TS, because 'more data means a better estimate.' Whose approach is correct, and why?

- A. Engineer A's, because using transactions after SCORING_TS would leak information that would not genuinely have been available at the moment of scoring, regardless of whether it improves the raw estimate's statistical precision
- B. Engineer B's, because a larger sample always produces a more statistically reliable frequency estimate
- C. Both are equally correct, since SCORING_TS only affects the target variable, not the features
- D. Neither is correct; frequency should always be computed over a fixed 30-day window regardless of SCORING_TS
- E. Engineer B's, because leakage only applies to the 'amount' feature, not to 'frequency'

22 The audit task in Lab 02 separately checks for (a) exact duplicate rows and (b) entities requiring resolution via reg_id. Why are these treated as two distinct problems rather than one?

- A. Entity resolution replaces the need to check for duplicate rows entirely
- B. Exact duplicate rows are identical records that should likely be removed outright, whereas entity resolution addresses genuinely distinct transaction rows that belong to the same underlying customer under different identifiers — collapsing them is about grouping, not deduplication
- C. They are actually the same problem, and the lab's separation into two tasks is only a stylistic choice
- D. Duplicate rows only occur in the amount column, while entity resolution only occurs in the txn_time column
- E. Duplicate rows are a data-leakage issue, while entity resolution is a class-imbalance issue

23 For a categorical feature with thousands of rare levels (e.g., a merchant ID), which combination of concerns is MOST relevant to choosing between one-hot encoding and (out-of-fold) target encoding?

- A. High-cardinality categorical features should always be dropped rather than encoded
- B. There is no meaningful difference between one-hot and target encoding for high-cardinality features
- C. One-hot encoding would explode the feature space into extremely sparse, mostly-empty columns for rare levels, while target encoding compresses the category into a single informative numeric column — but only if done out-of-fold to avoid the self-leakage risk described for target encoding
- D. One-hot encoding is always preferred for high-cardinality features because it avoids all forms of leakage entirely
- E. Target encoding cannot be combined with cross-validation under any circumstances

24 A fraud model reaches ROC-AUC 0.99 on a random (non-grouped) train/test split using: (i) a scaler fit on the full dataset before splitting, (ii) naive (non-out-of-fold) target encoding of `merchant_id`, and (iii) `manual_review_score` included as a feature. Which single change, made first, would you expect to cause the LARGEST drop in the reported score?

- A. Switching target encoding to out-of-fold — this is described as having the single largest quantified effect of the three issues
 - B. All three issues are documented as contributing exactly equal amounts to the inflated score
 - C. None of the three changes would affect the reported score, since ROC-AUC is insensitive to leakage by construction
 - D. Removing `manual_review_score` (and `settlement_status`) — the lab’s own headline result shows these two post-outcome columns alone account for roughly +0.37 of ROC-AUC, a larger single effect than either the scaler timing or the naive target encoding
 - E. Switching the scaler to fit only on the training split — this is described as having the single largest quantified effect of the three issues
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Topic 3: Classical ML Evaluation & Imbalanced Classes

PR-AUC vs ROC-AUC baselines, cost-matrix threshold reasoning, calibration direction, precise numeric F1/CV calculations

25 Model X achieves $\text{PR-AUC} = 0.35$ on the loan default data (~6% positive rate). A colleague claims this is weak because it is ‘far below 1.0.’ What is the deepest flaw in judging PR-AUC against an absolute scale like this?

- A. PR-AUC and ROC-AUC always have an identical 0.5 no-skill baseline, so 0.35 is indeed a poor score
 - B. PR-AUC cannot be meaningfully compared to any baseline at all
 - C. 0.35 is a poor score because PR-AUC should always exceed 0.90 to be considered acceptable, regardless of class balance
 - D. PR-AUC only applies to multi-class problems, so this comparison is invalid for a binary target
 - E. Unlike ROC-AUC, PR-AUC’s baseline (the score achieved by a random classifier) is not a fixed 0.5 — it is close to the positive class’s prevalence, so at ~6% positive rate a ‘no-skill’ baseline PR-AUC is around 0.06, meaning 0.35 actually reflects discrimination far above chance
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26 Under the lab's cost matrix (missed default costs $\sim 10\times$ a wrongly rejected good applicant), suppose the default rate were much higher — 40% instead of 6% — with the same relative cost ratio. What would you expect regarding the optimal threshold?

- A. The exact optimal threshold would likely shift from the lab's reported ~ 0.23 , since the optimal threshold depends jointly on the cost ratio AND the class balance/score distribution, not on the cost ratio alone
- B. The optimal threshold would remain exactly 0.23 regardless of class balance, since it is determined purely by the 10:1 cost ratio
- C. The optimal threshold would always equal the positive class rate exactly, i.e. 0.40
- D. The optimal threshold would necessarily rise to 0.50, the conventional default
- E. Changing the class balance has no effect on threshold selection, only on ROC-AUC

27 A model has excellent ROC-AUC (0.91) but its reliability diagram shows predicted probabilities of 0.8 corresponding to an observed default rate of only 0.5. What can be correctly concluded?

- A. This pattern indicates the model is underfitting the training data
- B. The model discriminates well (ranks defaulters above non-defaulters effectively) but is poorly calibrated (overconfident) — these are separate properties, and a model can have one without the other
- C. The model's ROC-AUC of 0.91 is impossible if it is poorly calibrated, so the reliability diagram must be computed incorrectly
- D. Poor calibration always implies poor ROC-AUC, so this scenario cannot occur
- E. The model should be retrained from scratch, since calibration issues cannot be fixed without retraining

28 In a dataset where both class imbalance (6% positive) AND customer-level grouping matter simultaneously, which statement correctly describes the tension between StratifiedKFold and GroupKFold?

- A. StratifiedKFold automatically respects any grouping variable present in the dataset without being told about it
- B. This tension can never be resolved, so cross-validation should be abandoned in favour of a single train/test split
- C. A plain StratifiedKFold does not respect customer grouping (risking the same customer appearing in both train and validation), while a plain GroupKFold does not guarantee each fold preserves the $\sim 6\%$ positive rate — satisfying both properties requires a stratified-group split strategy
- D. StratifiedKFold and GroupKFold are mathematically identical, so there is no real tension
- E. GroupKFold automatically preserves class balance in every fold as a built-in guarantee

29 Two models are compared: Model A has precision=0.90, recall=0.10; Model B has precision=0.50, recall=0.50. Which has the higher F1 score, and what does this reveal about F1's behaviour relative to a simple arithmetic average of precision and recall?

- A. Both models have identical F1 scores, since both have an arithmetic mean of 0.5
- B. F1 cannot be computed for Model A because its recall is below 0.5
- C. Model A has the higher F1 because F1 always favours whichever metric (precision or recall) is larger
- D. Model B has the higher F1 (0.50 vs. Model A's $F1 \approx 0.18$, despite Model A's arithmetic mean of precision and recall being 0.50, identical to Model B's) — F1 (the harmonic mean) punishes a large imbalance between precision and recall far more heavily than a simple average would
- E. Model A has the higher F1, since its precision of 0.9 dominates the score

30 A model's raw output scores are transformed by any strictly increasing function (e.g., converting logits to probabilities via sigmoid). What happens to the model's ROC-AUC as a result?

- A. ROC-AUC always increases after such a transform, since probabilities are more interpretable than raw logits
- B. ROC-AUC always decreases, since information is lost when compressing logits into a bounded $[0,1]$ range
- C. ROC-AUC becomes undefined, since it can only be computed on raw logits
- D. ROC-AUC changes unpredictably, since the transform alters the underlying data distribution
- E. ROC-AUC is unchanged, because it depends only on the relative ranking of scores across examples, and a strictly increasing transform preserves that ranking exactly

31 A lender wants to MINIMISE overall cost, and explicitly states missed defaults cost 10x more than wrongly rejected good applicants. Which metric, optimised directly, would most naturally align the model's threshold choice with this stated business objective?

- A. None of the standard fixed metrics (accuracy, F1, ROC-AUC, PR-AUC) directly encodes an asymmetric 10:1 cost ratio; the correct approach is to sweep the decision threshold against the explicit cost matrix itself, rather than optimise any single symmetric metric
- B. Accuracy, since it is the most intuitive metric for any business stakeholder
- C. ROC-AUC, since it is threshold-independent and therefore automatically accounts for any cost ratio
- D. F1 score, since its harmonic-mean structure automatically encodes a 10:1 cost asymmetry
- E. Precision alone, since minimising missed defaults is fundamentally a precision problem

32 With a 6% positive rate and StratifiedKFold with five folds, approximately how many positive (defaulting) examples would you expect in EACH validation fold, if the total dataset has 10,000 rows?

- A. It cannot be estimated without knowing the exact random seed used
 - B. Approximately 120 ($\approx 6\%$ of the 2,000 rows in each of the five folds)
 - C. Approximately 600 ($\approx 6\%$ of the full 10,000-row dataset, present entirely in every fold)
 - D. Approximately 24 ($\approx 6\%$ of a single row, divided across five folds)
 - E. Exactly 500, since five folds always split any dataset into equal positive/negative halves
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33 A reliability diagram shows the model's curve consistently BELOW the diagonal at high predicted probabilities (e.g., predicted 0.9 corresponds to an observed frequency of only 0.6). What does this specific pattern indicate, as opposed to the curve lying consistently ABOVE the diagonal?

- A. This pattern only occurs when StratifiedKFold has not been used
 - B. Curves below vs. above the diagonal are interchangeable descriptions of the same calibration phenomenon
 - C. The model is overconfident at high predicted probabilities — it predicts higher probabilities of default than are actually observed, the opposite pattern from underconfidence (curve above the diagonal, where predicted probabilities are lower than observed frequencies)
 - D. The model is underconfident at high predicted probabilities, identical to a curve lying above the diagonal
 - E. This pattern indicates the model has poor ROC-AUC, regardless of its discrimination ability
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34 A model is tuned and its threshold is set to 0.23 based on a cost matrix derived from CURRENT default rates and CURRENT recovery costs. Six months later, recovery costs for missed defaults double, while nothing else changes. What is the most defensible course of action?

- A. The entire model must be retrained from scratch, since threshold and model weights are the same thing
 - B. Increase the threshold to exactly 1.0, since higher costs always imply the strictest possible threshold
 - C. Switch metrics entirely from ROC-AUC to accuracy, since accuracy is more robust to changing costs
 - D. Recompute the cost matrix with the new recovery costs and re-sweep the threshold — the threshold is a function of the cost matrix, and a change in real-world costs should trigger a corresponding change in the decision threshold, not a full model retrain
 - E. No action is needed, since the threshold of 0.23 was fixed once during the original tuning and should never be revisited
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35 A model reports ROC-AUC = 0.63 (honest features) and, separately, PR-AUC = 0.30 on the same imbalanced ($\sim 6\%$) evaluation set. A stakeholder says: 'A PR-AUC of only 0.30 out of a possible 1.0 proves this model is far weaker than its ROC-AUC of 0.63 suggests.' What is the strongest, most precise rebuttal?

- A. The stakeholder is correct; any PR-AUC below 0.5 categorically indicates a weak or useless model, regardless of class balance
 - B. ROC-AUC and PR-AUC are mathematically identical for any dataset, so a discrepancy between them signals a computational bug
 - C. PR-AUC is deprecated and should never be reported alongside ROC-AUC
 - D. The comparison is invalid because ROC-AUC and PR-AUC can never both be computed on the same evaluation set
 - E. The two metrics are not on a comparable numeric scale: ROC-AUC's no-skill baseline is a fixed 0.5, while PR-AUC's no-skill baseline is close to the positive class's prevalence (~ 0.06 here); a PR-AUC of 0.30 against a ~ 0.06 baseline reflects roughly 5x lift over chance, a signal a direct 0.63-vs-0.30 comparison obscures
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Topic 4: Ensemble Methods & Hyperparameter Tuning

Correlated bootstraps, OOB limits, AdaBoost's exponential-loss weakness, TPE mechanics, Gini vs. permutation bias

36 If B independent, identically distributed base learners each have variance σ^2 , averaging their predictions gives variance approximately σ^2/B . Why does this simple formula OVERSTATE the actual variance reduction bagging achieves in practice?

- A. Bootstrap-sampled trees are not fully independent — they are trained on overlapping data drawn from the same training set and are therefore positively correlated, so the true variance of the average lies above σ^2/B , unlike the idealised fully-independent case
 - B. The formula understates the variance reduction, since bagged trees are always perfectly negatively correlated
 - C. The formula is exactly correct for bagging, with no caveats, because bootstrap samples are drawn independently by definition
 - D. Bagging does not reduce variance at all; only boosting does, so the formula is irrelevant
 - E. The formula only applies to boosting, not bagging, so it cannot be evaluated here
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37 For a bootstrap sample of size n drawn from a training set of size n , the exact probability a given row is left out is $(1 - 1/n)^n$. How does this probability behave as n grows from small (e.g. $n=10$) to very large (e.g. $n=100,000$)?

- A. It oscillates unpredictably and has no stable limiting value as n grows
 - B. It converges toward $\approx 36.8\%$ ($1/e$) from below as n grows, and is already close to that limit even for n in the low hundreds — the $\approx 36.8\%$ OOB rule of thumb applies essentially regardless of dataset size, once n is reasonably large
 - C. It converges toward 50% as n grows, matching the general intuition that half of any sample is always left out
 - D. It decreases toward 0% as n grows, meaning OOB evaluation becomes impossible for very large datasets
 - E. It increases toward 100% as n grows, meaning almost every row is eventually left out of every bootstrap sample
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38 AdaBoost's exponential loss is named as a specific weakness on datasets with mislabeled rows (e.g., disputed transactions), because such rows:

- A. Have no effect on AdaBoost, since row weights only apply to correctly labelled rows
 - B. Cause AdaBoost to fail to converge mathematically, producing an undefined model
 - C. Receive exponentially growing weight once the committee is confidently wrong about them, causing later boosting stages to distort the decision boundary around these mislabeled points specifically to accommodate them
 - D. Are automatically dropped from the training set by AdaBoost's row-weighting mechanism
 - E. Receive exponentially SHRINKING weight, causing AdaBoost to ignore them entirely, which is actually beneficial
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39 Which statement correctly describes the relationship between AdaBoost and modern gradient boosting, rather than treating them as unrelated algorithms?

- A. Gradient boosting was developed before AdaBoost and AdaBoost is a later refinement of it
 - B. AdaBoost only works for regression problems, while gradient boosting only works for classification
 - C. AdaBoost and XGBoost are mathematically identical, differing only in software implementation
 - D. AdaBoost is a historical special case that expresses the same underlying idea — sequentially correcting what the committee still gets wrong — through row reweighting rather than the residual-fitting view used by gradient boosting, XGBoost, and LightGBM
 - E. AdaBoost and gradient boosting share no conceptual relationship; AdaBoost is a bagging method while gradient boosting is a boosting method
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40 A colleague suggests: 'We can get the same decorrelation benefit as a random forest's feature subsampling simply by training more trees on more bootstrap samples, without ever restricting the features considered at each split.' Is this reasoning sound?

- A. Yes — bootstrap sampling of rows and feature subsampling at each split achieve mathematically identical decorrelation effects, so either alone is sufficient
- B. Yes — training more trees always eliminates correlation between trees regardless of how splits are chosen
- C. No — but only because more trees increase training time, not because of any decorrelation limitation
- D. The question is moot, since random forests do not use bootstrap sampling of rows at all
- E. No — more bootstrap samples alone still allow every tree to pick the single strongest feature at the top split whenever that feature is highly informative, keeping the trees correlated; feature subsampling at each split forces different trees to rely on different features, which row-bootstrapping alone does not achieve

41 Student A says: 'TPE models $p(\text{hyperparameters} \mid \text{good performance})$ and $p(\text{hyperparameters} \mid \text{bad performance})$ separately from past trials, then samples where the ratio favours good performance.' Student B says: 'TPE evaluates every point on a fixed grid, just like grid search, but in a randomized order.' Which is correct?

- A. Student A — TPE is a sequential, model-based (Bayesian-style) search that learns from the distribution of past good vs. bad trials, which is precisely why it is more sample-efficient than grid or random search; Student B's description is closer to grid/random search
- B. Student B — TPE is fundamentally a randomized grid search, and Student A's description mistakenly applies to a different algorithm entirely
- C. Both students are equally correct, since TPE combines exhaustive grid evaluation with probabilistic modelling
- D. Neither student is correct; TPE requires no modelling of past trials whatsoever
- E. The distinction is purely semantic and has no practical implication for how many trials are needed

42 A MedianPruner stops a trial early if its intermediate performance falls below the median of previous trials at the same point in training. What is the main practical benefit of this, distinct from simply running fewer total trials?

- A. It increases the total number of trials run, since pruned trials are immediately replaced by two new ones
- B. It reallocates compute away from trials that are already underperforming relative to peers at the same stage, letting the search spend its fixed total budget on more promising configurations rather than fully training every single one to completion
- C. It guarantees the globally optimal hyperparameter configuration will always be found within 50 trials
- D. It eliminates the need for a validation metric altogether, since pruning replaces evaluation
- E. It only prunes trials that have already fully completed training, so it has no effect on wall-clock time

43 For a given tabular dataset, built-in Gini importance ranks feature X as the single most important, while permutation importance (on held-out data) ranks feature X as one of the least important. What is the most likely explanation?

- A. Feature X must be the target variable itself, since only the target can show this pattern
 - B. This indicates the model has zero predictive power for any feature
 - C. Feature X likely has high cardinality (e.g., many unique values), which biases Gini importance upward regardless of X's true predictive value, while permutation importance — measured on held-out data by observing the actual performance drop when X is shuffled — is not subject to this cardinality bias
 - D. Permutation importance must have been computed incorrectly, since Gini importance is always the more trustworthy of the two
 - E. This discrepancy is impossible; Gini and permutation importance always agree exactly by construction
-

44 Which pair correctly distinguishes the ROLE of subsample and colsample_bytree from the role of reg_alpha and reg_lambda?

- A. All four hyperparameters control exactly the same thing: the learning rate schedule
 - B. subsample and colsample_bytree only apply to random forests, never to gradient boosting
 - C. reg_alpha and reg_lambda control the number of estimators, while subsample and colsample_bytree control tree depth
 - D. subsample and colsample_bytree control STOCHASTICITY — the fraction of rows/columns randomly used per boosting round, akin to bagging-style randomness — while reg_alpha and reg_lambda are explicit L1/L2 PENALTY terms added directly to the boosting objective
 - E. subsample and colsample_bytree are L1/L2 penalty terms, while reg_alpha and reg_lambda control row/column sampling randomness
-

45 A random forest reports oob_score_ = 0.87 (accuracy). The same model, evaluated on a separately held-out test set collected one year after the training data, scores only 0.71. Which explanation is MOST consistent with 'OOB is not out-of-time'?

- A. The OOB score of 0.87 must be a computational error, since OOB and true held-out test performance are mathematically guaranteed to be identical
 - B. The gap is fully explained by the random forest using bootstrap sampling, which always overstates test performance by exactly this margin
 - C. The gap indicates the test set must have a different set of features than the training set
 - D. OOB scores are always lower than true test performance, so an 0.87-to-0.71 drop is actually the wrong direction and indicates leakage
 - E. The OOB estimate randomises over rows drawn from the same time period as training and cannot detect distributional drift that occurred over the intervening year; the drop to 0.71 likely reflects real-world drift that a row-wise OOB estimate is structurally unable to surface
-

46 A team bags very SHALLOW decision trees (`max_depth=2`) instead of deep trees. Based on bagging's characterisation as a variance-reduction technique, what limitation would you expect?

- A. Shallow trees individually have high bias (too simple to capture complex patterns), and bagging only reduces variance, not bias — bagging many shallow, high-bias trees still leaves the ensemble with a high-bias 'floor' averaging alone cannot fix; deep, low-bias/high-variance trees are the better base learner for bagging
 - B. Bagging shallow trees is strictly superior to bagging deep trees in every case, since shallow trees never overfit
 - C. Bagging automatically converts high-bias base learners into low-bias ones through the averaging process
 - D. The choice of tree depth has no effect whatsoever on a bagged ensemble's performance
 - E. Shallow trees should never be used as an ensemble's base learner under any ensemble method, bagging or boosting
-

Topic 5: Unsupervised Learning & Dimensionality Reduction

Inertia monotonicity, DBSCAN eps direction, PCA numeric reasoning, t-SNE sensitivity, hierarchical clustering, negative silhouette

47 As k increases from 1 upward, K-Means inertia (within-cluster sum of squares) computed on the SAME data will:

- A. Behave unpredictably with no general relationship to k
 - B. Monotonically decrease (or stay the same) as k increases, reaching zero when k equals the number of data points — this is exactly why the elbow method looks for a bend rather than a minimum, since inertia has no genuine interior minimum to find
 - C. Monotonically increase as k increases, which is why the elbow method looks for a maximum
 - D. Stay exactly constant regardless of k , since inertia only depends on the number of features
 - E. Decrease and then increase again, forming a U-shape with a genuine global minimum at some interior k
-

48 The elbow method suggests $k=3$ (a visible bend in inertia), but the silhouette score is actually higher at $k=5$ than at $k=3$. What is the most defensible interpretation?

- A. This disagreement proves that K-Means should never be used on this dataset under any k
 - B. Inertia and silhouette score are actually the exact same quantity expressed in different units, so a disagreement is impossible
 - C. The two diagnostics can legitimately disagree, since they measure different things (inertia's rate of decrease vs. relative cluster separation/cohesion); this disagreement is a signal to examine both $k=3$ and $k=5$ solutions substantively rather than mechanically trusting one diagnostic over the other
 - D. The silhouette score must be computed incorrectly whenever it disagrees with the elbow method, since the two are mathematically guaranteed to agree
 - E. The elbow method should always be trusted over the silhouette score in any disagreement
-

49 In DBSCAN, if ϵ is set far too SMALL for the data's actual density, what is the most likely outcome?

- A. DBSCAN will behave identically to K-Means with k equal to the number of points
 - B. DBSCAN will raise a runtime error and fail to produce any output
 - C. An ϵ that is too small has no effect on the clustering result, only on runtime
 - D. Most points fail to have enough neighbours within the tiny radius to be classified as core points, so most of the dataset ends up labelled as noise (-1), producing many tiny or empty clusters rather than a few meaningful ones
 - E. All points will be merged into a single giant cluster, since a smaller radius always merges more points together
-

50 A colleague says: 'PCA and simply dropping the least-correlated-with-target features are basically the same dimensionality reduction technique.' What is the key conceptual error here?

- A. There is no error; PCA and target-correlation-based feature dropping are mathematically identical procedures
 - B. PCA always produces fewer features than target-based feature selection, which is the only real difference
 - C. PCA is a supervised technique that explicitly uses the target variable, just like target-correlation-based selection
 - D. Feature selection based on target correlation always outperforms PCA, so the two should never be compared
 - E. PCA constructs NEW features as linear combinations (orthogonal directions of maximum variance) of ALL original features, without ever consulting the target variable, whereas dropping low-target-correlation features is a target-aware SELECTION among the ORIGINAL features — these are fundamentally different operations (unsupervised transformation vs. supervised selection)
-

51 A covariance matrix has eigenvalues 8, 4, 2, 1 (summing to 15, the total variance). What is the explained variance ratio of the first TWO principal components, to the nearest percent?

- A. 80% — $(8+4)/15 \approx 0.80$
 - B. 53% — $8/15 \approx 0.53$, counting only the single largest eigenvalue
 - C. 93% — $(8+4+2)/15 \approx 0.93$, incorrectly including the third eigenvalue
 - D. 27% — $4/15 \approx 0.27$, counting only the second eigenvalue alone
 - E. 100%, since any two of four eigenvalues are assumed to explain all the variance
-

52 A student runs t-SNE twice on the identical dataset with two very different 'perplexity' values and gets visibly different cluster shapes and apparent inter-cluster distances both times. What is the correct interpretation?

- A. This sensitivity only affects categorical data, not the type of numeric feature set used in this course's labs
 - B. This is expected and reinforces the course's warning: t-SNE's axes and inter-cluster distances are not intrinsically meaningful and can shift substantially with hyperparameters like perplexity, which is precisely why t-SNE should be used only for exploratory visualisation, never as a feature-engineering step feeding a downstream model
 - C. This indicates a bug, since t-SNE is mathematically guaranteed to produce an identical embedding regardless of perplexity
 - D. This proves PCA should never be used either, since it has the same sensitivity to hyperparameters as t-SNE
 - E. The correct perplexity can always be uniquely determined from the data with no ambiguity, so this discrepancy should not occur in practice
-

53 The unsupervised chapter uses reconstruction error from a PCA projection as a candidate anomaly signal, flagging points above the 99.5th percentile of error. What does a HIGH PCA reconstruction error for a given point most directly indicate?

- A. The point has zero variance across all retained principal components
 - B. The point is guaranteed to be a duplicate row
 - C. The point's structure is poorly captured by the retained principal components — it deviates substantially from the low-dimensional 'manifold' that best describes the bulk of the data — which is why unusually high reconstruction error is used as a candidate anomaly signal rather than a definitive fraud label
 - D. The point has an unusually LOW value on every original feature simultaneously
 - E. The point was definitely mislabeled in the dataset and should be removed before any further analysis
-

54 Unlike K-Means, which requires specifying k in advance and re-running the whole algorithm for each candidate k , agglomerative hierarchical clustering with a chosen linkage criterion:

- A. Cannot produce more than two clusters under any circumstances
 - B. Requires the data to be reduced via PCA before it can be applied at all
 - C. Always produces identical results to DBSCAN regardless of the linkage criterion chosen
 - D. Builds a single dendrogram (merge tree) in one pass that can be 'cut' at different heights to obtain different numbers of clusters after the fact, without needing to rerun the clustering algorithm for each candidate k
 - E. Also requires k to be fixed in advance and re-run separately for every candidate k , identically to K-Means
-

55 A specific data point has a NEGATIVE silhouette value. What does this most directly suggest about that point?

- A. The point is a perfect representative of its assigned cluster's centroid
 - B. The point has a negative value on every one of its original features
 - C. A negative silhouette value is mathematically impossible, so this indicates a computational error
 - D. The point must be an outlier that DBSCAN would also label as noise, since silhouette and DBSCAN's noise label are computed identically
 - E. The point is, on average, closer to points in a DIFFERENT cluster than to points in its OWN assigned cluster — a sign that it may have been assigned to the wrong cluster
-

56 Increasing DBSCAN's `min_samples` parameter (while holding `eps` fixed) tends to:

- A. Make the density requirement for a core point stricter, generally producing fewer, denser clusters and reclassifying more borderline points as noise
 - B. Have no effect on the clustering result, since `min_samples` only affects runtime, not cluster assignment
 - C. Always increase the total number of clusters found, regardless of the data
 - D. Convert DBSCAN into an algorithm mathematically identical to K-Means
 - E. Only affect points that are already labelled as noise, never core or border points
-

57 A team runs PCA to 2 components purely for visualisation, then visually identifies what look like 4 well-separated blobs in the 2D PCA scatterplot, and proposes deploying 'PCA-component-1 > threshold' as a production fraud rule based on this visual separation. What is the strongest combined objection?

- A. The only issue is that PCA components must be renamed before they can be used in a business rule
 - B. Two PCA components retained purely for visualisation may capture only a small fraction of total variance (far less than the 95% threshold used for modelling), so apparent visual separation in 2D may not reflect genuine structure in the full feature space; moreover, even genuine clusters are not labels — any proposed rule still needs proper supervised evaluation before deployment
 - C. There is no objection; visually separated blobs in a PCA scatterplot are definitionally equivalent to a validated, deployable fraud rule
 - D. The only issue is that PCA should never be used for visualisation, only for feature reduction feeding a supervised model
 - E. The only issue is that 4 clusters is too many; a production rule should never use more than 2 clusters
-

Topic 6: Deep Learning Fundamentals (MLP)

Dropout train/inference behaviour, sigmoid derivative bound, Adam's two moments, softmax vs sigmoid mismatch, underfitting vs overfitting traps

58 During TRAINING, a Dropout(0.3) layer randomly zeroes 30% of its input units on each forward pass. What happens to this layer's behaviour at INFERENCE (prediction) time in a standard implementation, and why does this matter?

- A. Dropout has no effect on the network at any stage; it is purely a training-time visualisation tool
 - B. Dropout is applied only to the input layer, never to hidden layers like the ones in Lab 06's MLP
 - C. At inference time, dropout is turned off (no units are zeroed), and the layer's outputs are typically scaled to account for the change (inverted dropout), so predictions use the FULL, deterministic network — a model's inference-time behaviour is NOT literally 'run with 30% of units randomly missing' the way it was during training
 - D. Dropout continues to randomly zero 30% of units at inference time as well, producing a different random prediction every time the same input is scored
 - E. Dropout permanently removes 30% of the network's weights after training completes, shrinking the model file
-

59 The sigmoid function's derivative, $\sigma'(z) = \sigma(z)(1-\sigma(z))$, has a maximum value of exactly 0.25. Why is this specific numeric fact relevant to the vanishing gradient problem in a DEEP sigmoid network?

- A. A maximum derivative of 0.25 means sigmoid derivatives always EXCEED 1, which would cause EXPLODING rather than vanishing gradients
- B. This fact only applies to convolutional networks, never to fully-connected MLPs
- C. This bound is only relevant to networks with fewer than 3 layers, and is irrelevant to genuinely deep networks
- D. Because backpropagation multiplies these derivatives together across layers via the chain rule, and even the best-case derivative of 0.25 per layer causes the gradient magnitude to shrink geometrically (roughly 0.25^L after L layers) as it propagates backward, which is why deep sigmoid networks are especially prone to vanishing gradients
- E. This fact is irrelevant, since the vanishing gradient problem is caused exclusively by the choice of loss function, not the activation function's derivative

60 Adam's update rule uses a first-moment estimate (m , an exponential moving average of the gradient) and a second-moment estimate (v , an exponential moving average of the squared gradient). Which statement correctly distinguishes their roles?

- A. The first moment scales the step size per parameter, while the second moment provides momentum-like directional smoothing — the reverse of Adam's actual design
- B. Both moments serve an identical purpose and are averaged together into a single quantity before being used
- C. The first moment is only used during the first training epoch, and the second moment is only used afterward
- D. Neither moment affects the learning rate; both are used solely for logging and monitoring purposes
- E. The first moment (m) provides momentum-like smoothing of the gradient's direction across updates, while the second moment (v) is used to adaptively SCALE the per-parameter step size — parameters with historically large squared gradients get smaller effective steps, and vice versa

61 A team building a 3-class loan-purpose classifier ('auto', 'education', 'business') mistakenly uses a single sigmoid output unit with `binary_crossentropy` loss, as if it were the binary default-prediction problem from Lab 06. What is the most direct consequence of this mismatch?

- A.** A single sigmoid unit can only represent a probability for ONE binary outcome and cannot represent a valid probability distribution over three mutually exclusive classes; the correct setup for K mutually exclusive classes is K output units with a softmax activation and `categorical_crossentropy` loss
 - B.** There is no consequence; `binary_crossentropy` with a single sigmoid unit is mathematically equivalent to softmax with three units for any number of classes
 - C.** The mismatch only affects training speed, not the validity of the model's predictions
 - D.** Sigmoid and softmax are interchangeable for any number of output classes, so this is a stylistic choice only
 - E.** This setup is actually required whenever the three classes are imbalanced in frequency
-

62 A team sets early stopping's patience parameter to 1 (stop immediately if validation loss fails to improve for even a single epoch) instead of the lab's `patience=10`. What is the most likely downside of this overly aggressive setting?

- A.** Patience values below 5 are not supported by Keras and would raise an error
 - B.** Because validation loss can fluctuate noisily from epoch to epoch even while the true underlying trend is still improving, a patience of 1 risks stopping training prematurely on a single noisy uptick, before the model has genuinely converged — `patience=10` exists specifically to tolerate this kind of short-term noise
 - C.** A patience of 1 has no downside and is strictly preferable to `patience=10` in every scenario, since it always saves training time with no cost to model quality
 - D.** Patience only affects the dropout rate, not when training stops, so this change would have no effect at all
 - E.** A low patience value causes the model to systematically overfit MORE than a high patience value would
-

63 Class imbalance can be addressed either by passing `class_weight` to `model.fit()` OR by oversampling the minority class with SMOTE before training. Which statement correctly distinguishes these two approaches?

- A. SMOTE reweights the loss function, while `class_weight` synthesises new synthetic rows
 - B. Neither technique has any effect on a neural network; both are exclusively for tree-based models
 - C. `class_weight` leaves the training data itself unchanged and instead reweights each example's contribution to the LOSS function during optimisation, whereas SMOTE physically synthesises new minority-class training examples, changing the composition of the dataset the model actually sees
 - D. `class_weight` and SMOTE are two names for exactly the same underlying mechanism, differing only in which library implements them
 - E. `class_weight` physically duplicates minority-class rows in the dataset, identical to naive oversampling
-

64 The course's 'honest conclusion' states the MLP reaches PR-AUC roughly comparable to the tuned gradient-boosted model but takes far longer to train. A student concludes: 'Therefore, gradient boosting is always faster to train than any neural network, on any dataset.' What is the flaw in this generalisation?

- A. Gradient boosting is actually always slower than neural networks in every documented case in this course
 - B. Training time comparisons are meaningless and were never discussed in the course material
 - C. The comparison only applies to convolutional networks, not MLPs, so the student's claim is unrelated to Lab 06 entirely
 - D. The comparison in the course material is specific to THIS tabular loan dataset and THIS particular architecture/tuning setup; the general training-time relationship between gradient boosting and neural networks depends heavily on dataset size, data type (tabular vs. unstructured), hardware, and architecture choice — it is an empirical observation on one case, not a universal law
 - E. The student's generalisation is entirely correct and is stated as a universal law in the course material, with no caveats
-

65 Both L2 weight decay and dropout are used together in Lab 06's regularisation stack. Which statement correctly distinguishes HOW each mechanism reduces overfitting?

- A. L2 weight decay and dropout are mathematically identical operations, so combining them provides no additional benefit over using either alone
- B. Dropout adds a penalty term to the loss based on squared weights, while L2 weight decay randomly deactivates units — the reverse of their actual roles
- C. Both mechanisms work exclusively by reducing the number of training epochs
- D. L2 weight decay only applies to convolutional layers, never to Dense layers like those in the Lab 06 MLP
- E. L2 weight decay adds a penalty term to the loss proportional to the sum of squared weights, directly discouraging any individual weight from growing large; dropout instead randomly deactivates units during training, forcing the network to not rely too heavily on any single unit or co-adapted group of units — the two act through different, complementary mechanisms

66 Why would using ReLU (instead of sigmoid) as the OUTPUT layer's activation function be problematic for Lab 06's binary default-prediction task, even though ReLU is preferred for the HIDDEN layers?

- A. ReLU's output range is $[0, \infty)$, which cannot represent a valid probability (which must lie in $[0, 1]$); `binary_crossentropy` loss expects a probability-like output, so the output layer specifically needs an activation like sigmoid that is bounded to $[0, 1]$, even though ReLU's unbounded, non-saturating behaviour is exactly what makes it preferable inside the hidden layers
- B. There is no problem; ReLU is equally suitable for both hidden and output layers in every binary classification task, including this one
- C. ReLU would work fine for the output layer as long as the hidden layers use sigmoid instead
- D. The output layer's activation choice has no effect on what loss function can be used
- E. Sigmoid should actually be used in the hidden layers instead of ReLU, and ReLU reserved only for the output layer

67 Consistent with the course's leakage-safe practice seen elsewhere, the StandardScaler used to preprocess inputs to the Lab 06 MLP should be:

- A. Refit on the test set specifically, since the test set is the most representative of real deployment conditions
- B. Fit once on the training split only, and that SAME fitted scaler (with its stored mean/std) reused to transform validation data, test data, and any new data at serving time — never refit on validation, test, or production data
- C. Refit independently every time new data needs to be scored in production, using only that new data's own mean and standard deviation
- D. Fit jointly on the combined training and validation data, since both are used somewhere in the modelling process
- E. Not needed at all for neural networks, since Dense layers automatically standardise their own inputs internally

68 A student trains the Lab 06 MLP with dropout, L2 weight decay, and early stopping (patience=10) all active, yet training and validation loss remain almost identical and both plateau at a relatively HIGH loss value very early in training (e.g., by epoch 5), with neither improving substantially afterward. What is the most likely underlying issue, and why do the regularisation settings NOT explain it?

- A. This pattern can only be explained by a bug in the early stopping callback's patience parameter
- B. This pattern indicates the class_weight parameter must be set incorrectly, since only class imbalance can cause matched train/validation loss
- C. This pattern (high, closely-matched train/validation loss that plateaus early) is a sign of UNDERFITTING, not overfitting — the regularisation stack exists specifically to combat overfitting (a gap between train and validation performance), and none of these mechanisms explain or fix a model too constrained or too simple to fit the training data well in the first place
- D. This is the expected and desirable signature of well-functioning regularisation, and no further investigation or changes are needed
- E. This pattern indicates severe overfitting, and the fix is to increase dropout and weight decay even further

Topic 7: Convolutional Neural Networks & Computer Vision

Output-size and receptive-field arithmetic, parameter counting, frozen-base gradient flow, flip-vs-rotation reasoning, confusion-matrix cost framing

69 For an input of size 64x64, a kernel size of 5, 'valid' padding (p=0), and a stride of 2, what is the output spatial size, using $\text{output} = \text{floor}((n + 2p - f)/s) + 1$?

- A. 60x60 — obtained by incorrectly using stride 1 in the formula instead of the given stride of 2
- B. 59x59 — obtained by omitting the '+1' term at the end of the formula
- C. 128x128 — obtained by incorrectly multiplying instead of dividing by the stride
- D. 30x30 — $\text{floor}((64 + 0 - 5)/2) + 1 = \text{floor}(59/2) + 1 = 29 + 1 = 30$
- E. 32x32 — obtained by incorrectly dividing 64 by the stride alone, ignoring the kernel size

70 Two consecutive Conv2D(3x3, stride=1) layers, with no pooling in between, are stacked. What is the effective receptive field, in terms of the ORIGINAL input pixels, of a single unit in the output of the SECOND layer?

- A. 3x3 — stacking layers has no effect on receptive field size, which always equals the kernel size of the LAST layer only
- B. 6x6 — obtained by simply adding the two kernel sizes together (3+3)
- C. 9x9 — obtained by multiplying the two kernel sizes together (3x3)
- D. 1x1 — obtained by assuming receptive field shrinks with each additional layer
- E. 5x5 — each 3x3 layer adds (kernel_size - 1) = 2 pixels to the receptive field per layer, so two stacked 3x3 layers give a receptive field of 3 + 2 = 5 pixels per side

71 A Conv2D layer has 32 filters, each of size 3x3, applied to an input with 3 channels (e.g., an RGB image), and includes a bias term per filter. How many trainable parameters does this single layer have?

- A. 896 — $(3 \times 3 \times 3 + 1 \text{ bias}) \times 32 \text{ filters} = (27+1) \times 32 = 28 \times 32 = 896$
 - B. 288 — obtained by omitting the bias term and the input channel count: $3 \times 3 \times 32$
 - C. 27 — obtained by computing only the kernel size times input channels for a SINGLE filter: $3 \times 3 \times 3$
 - D. 9,216 — obtained by incorrectly squaring the correct parameter count
 - E. 32 — obtained by counting only the number of filters, ignoring kernel size and channels entirely
-

72 A feature map of shape 14x14x256 passes through MaxPooling2D(2x2) versus, alternatively, through GlobalAveragePooling2D. What are the respective output shapes?

- A. Both operations discard the channel dimension entirely, producing a single scalar output
 - B. MaxPooling2D(2x2) produces 7x7x256 (halving each spatial dimension, channel count unchanged); GlobalAveragePooling2D produces a flat vector of shape 256 (collapsing the ENTIRE spatial extent per channel to a single average value)
 - C. Both operations produce an identical output shape of 7x7x256
 - D. MaxPooling2D(2x2) produces a flat vector of 256, while GlobalAveragePooling2D produces 7x7x256 — the reverse of their actual behaviour
 - E. MaxPooling2D(2x2) produces 28x28x256 (doubling spatial dimensions), while GlobalAveragePooling2D produces 14x14x1
-

73 When the convolutional base of MobileNetV2 is 'frozen' (as in the first stage of Lab 07's transfer learning), what precisely happens during backpropagation with respect to the frozen layers?

- A. Freezing only affects the forward pass, having no relationship to backpropagation or the optimiser at all
 - B. Freezing converts the layer's weights to zero, effectively removing it from the network
 - C. Gradients still flow BACKWARD through the frozen layers (as required to reach any trainable layers before them), but the frozen layers' own WEIGHTS are not updated by the optimiser — 'frozen' refers to weight updates being disabled, not to gradient computation being skipped entirely
 - D. Gradients are not computed at all for the frozen layers, and backpropagation stops entirely at the boundary between the frozen base and the new head
 - E. Freezing a layer means the optimiser continues to update its weights, but the model discards those updates before saving
-

74 Lab 07 avoids horizontal flips for anatomical reasons, but DOES use small rotations (up to 10°) and zoom. A student argues: 'If flips are disallowed because they create anatomically impossible images, rotation should also be disallowed, since a rotated X-ray isn't how the patient was actually positioned either.' What is the strongest distinction that justifies treating them differently?

- A. Rotation is disallowed and flips are allowed in Lab 07, the reverse of what was actually described
- B. The distinction is purely about computational cost: flips are more expensive to compute than rotations
- C. Small rotations are only used for colour images, so this augmentation is irrelevant to greyscale chest X-rays
- D. A small rotation ($\leq 10^\circ$) preserves the fundamental left-right anatomical relationship (heart still on the correct side relative to the rest of the anatomy) while plausibly modelling minor real-world variation in patient positioning during imaging; a horizontal FLIP, by contrast, categorically swaps left and right — the two differ in KIND (a structural violation vs. a bounded, realistic perturbation), not merely in degree
- E. There is no meaningful distinction; the student's argument is entirely correct and rotation should also be disallowed

75 A CNN uses THREE consecutive Conv2D(3x3, 'valid' padding, stride 1) layers with no pooling in between, on a 32x32 input. What is the spatial size after all three layers, and what does this illustrate about repeated 'valid' convolutions?

- A. 32x32 — 'valid' padding always preserves spatial size regardless of how many layers are stacked
- B. 20x20 — obtained by shrinking by the full kernel size (3) at each of the three layers instead of by (kernel_size - 1)
- C. 8x8 — obtained by incorrectly treating each 3x3 'valid' convolution as a pooling operation that quarters the spatial size
- D. 29x29 — obtained by only accounting for one of the three convolutional layers' shrinkage
- E. 26x26 — each 'valid' 3x3 convolution shrinks the spatial size by 2 (32→30→28→26), illustrating that stacking multiple 'valid'-padded convolutions causes CUMULATIVE shrinkage, which can eventually make further convolutions or pooling infeasible if not accounted for

76 Lab 07's fine-tuning stage uses a learning rate of $1e-5$, specifically much smaller than the rate used to train the new head from scratch. Which of the following best explains the DIRECTION of this difference?

- A. The pretrained convolutional weights already encode useful, well-trained representations, so large gradient updates risk overwriting that prior knowledge; a small learning rate allows gentle, incremental adjustment without destroying what was already learned, whereas the newly-initialised head has no prior knowledge to protect and can tolerate larger updates
- B. A smaller learning rate is used purely to make training run faster in wall-clock time
- C. A smaller learning rate is required because MobileNetV2 has fewer parameters than the new classification head
- D. The learning rate has no relationship to whether weights are pretrained or randomly initialised; the difference is arbitrary
- E. A larger learning rate would be used for fine-tuning specifically because the pre-trained weights are already known to be suboptimal for any new task

77 A chest X-ray classifier's confusion matrix on a 200-image test set (100 NORMAL, 100 PNEUMONIA) is: TP=85, FN=15, TN=95, FP=5. A colleague argues the model is 'basically fine' because overall accuracy is $(85+95)/200 = 90\%$. What is the sharpest, course-grounded critique of resting on this single number?

- A. The confusion matrix numbers given are internally inconsistent and cannot represent a real classifier
- B. The 90% accuracy figure treats the 15 false negatives (missed pneumonia cases) and the 5 false positives (unnecessary follow-ups) as equally costly, when the course explicitly frames a missed pneumonia case as far more clinically costly; the more clinically relevant number here is the 15% false-negative rate (15 missed cases out of 100 true PNEUMONIA cases), which a single aggregate accuracy figure obscures entirely
- C. The accuracy calculation itself is arithmetically wrong; it should be $(85+95)/(85+15+95+5)$ which is a different number entirely
- D. There is no valid critique; a 90% accuracy is unambiguously excellent regardless of the underlying confusion matrix breakdown
- E. The critique should focus exclusively on the 5 false positives, since false positives are always more clinically costly than false negatives in any medical context

78 Lab 07 states that augmentation 'matters more than architecture' because the training set is tiny. What is the precise mechanism by which augmentation helps in this specific situation, as opposed to simply architecture choice?

- A. Augmentation works by removing images from the training set that are judged to be low quality
- B. Augmentation's only benefit is reducing the total training time, unrelated to overfitting or generalisation
- C. Augmentation exposes the model to many plausible variations (rotations, shifts, zooms) of the SAME small set of original images during training, which helps the model learn features that are robust to those variations and reduces the degree to which it can simply memorise the small number of exact original training images
- D. Augmentation increases the actual number of unique patients in the dataset, directly solving the small-sample-size problem
- E. Augmentation and architecture choice are functionally identical operations, so the distinction drawn in the lab is not meaningful

79 A CNN consists of 5 blocks, each block being one Conv2D('same' padding, stride 1) followed by one MaxPooling2D(2x2, stride 2). Starting from a 256x256 input, what is the spatial size after all 5 blocks, and what is the network's total stride at this point?

- A. 1x1 spatial size, with a total stride of 256, since both the convolution AND the pooling layers are assumed to each halve the spatial dimensions
- B. 16x16 spatial size, with a total stride of 16, obtained by only counting 4 of the 5 pooling operations
- C. 8x8 spatial size, with a total stride of 5, incorrectly equating 'total stride' with 'number of blocks' rather than the cumulative downsampling factor
- D. 8x8 spatial size, with a total stride of 32 — 'same' padding keeps the Conv2D layers spatial-size-neutral, so only the five stride-2 pooling operations reduce resolution: $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8$, a cumulative factor of $2^5 = 32$
- E. 256x256 spatial size (unchanged), with a total stride of 1, since 'same' padding preserves spatial dimensions throughout the entire network including the pooling layers

Topic 8: Sequence Models: RNNs, LSTMs & Time-Series Forecasting

Windowing/split arithmetic, forget vs. input gate, parameter sharing across time, aggregation-before-differencing order, exploding gradients

80 With LOOKBACK=30 and a cleaned daily series of exactly 400 days, how many (X, y) supervised windowed examples can be constructed in total, before any train/test split is applied?

- A. 400 — one window can be formed for every single day in the series, regardless of the lookback length
- B. 30 — the number of examples equals the lookback length itself
- C. 13 — obtained by dividing 400 by 30 and rounding down, as if the windows did not overlap
- D. 430 — obtained by incorrectly adding the lookback length to the series length instead of subtracting it
- E. 370 — with a 30-day lookback window and a 400-day series, the first possible window uses days 1-30 to predict day 31, and the last uses days 370-399 to predict day 400, yielding $400 - 30 = 370$ total windows

81 Continuing from a 400-day series, LOOKBACK=30, with the last 60 days held out as test: how many of the 370 total windowed examples fall ENTIRELY within the test period (both the 30-day input window AND the target day among the last 60 days)?

- A. 30 — the test period has 60 days; a window entirely within it needs its target day among the last 60 AND its full 30-day lookback also within that same 60-day span, which is only possible starting from the 31st day of the 60-day test window onward, leaving $60 - 30 = 30$ valid windows
- B. 60 — one window for every day in the 60-day test period, without regard for whether its 30-day lookback fits entirely inside the test period
- C. 370 — all windows are assumed to fall within the test period regardless of the split boundary
- D. 0 — it is claimed no window can ever fall entirely within the test period under this scheme
- E. 90 — obtained by incorrectly adding the lookback length (30) to the test period length (60) instead of subtracting

82 The naive 'tomorrow equals today' baseline is 'surprisingly hard to beat on a smoothed series.' Why does applying a 7-day rolling average SPECIFICALLY make this baseline harder to beat, compared to forecasting the raw, unsmoothed daily series?

- A. Smoothing exclusively affects the LSTM's performance, having no relationship to the naive baseline's difficulty
- B. Smoothing removes much of the day-to-day noise/volatility from the series, so consecutive days' values become more similar to each other by construction; this makes 'predict yesterday's (already-smoothed) value' a naturally strong prediction for today's (also-smoothed) value, raising the bar any genuinely skilful model must clear
- C. Smoothing has no effect on how easy or hard the naive baseline is to beat; the statement refers only to the raw, unsmoothed series
- D. Smoothing makes the series MORE volatile day-to-day, which is why the naive baseline becomes weaker and easier to beat after smoothing
- E. The naive baseline can only be computed at all if the series has first been smoothed; it is undefined on raw daily data

83 Within an LSTM cell, the FORGET gate and the INPUT gate serve related but distinct purposes. Which statement correctly distinguishes them?

- A. The forget gate operates only during inference, while the input gate operates only during training
 - B. Both gates exclusively control the OUTPUT of the LSTM cell, having no relationship to the cell state itself
 - C. The forget gate controls how much of the EXISTING cell state to retain versus discard at this timestep, while the input gate controls how much of the NEW candidate information gets written into the cell state — one governs what's kept from the past, the other governs what's newly added
 - D. The forget gate controls what NEW information is added to the cell state, while the input gate controls what EXISTING information is discarded — the reverse of their actual roles
 - E. The forget gate and input gate are two names for the same gate, differing only by which deep learning library's terminology is used
-

84 A student claims: 'A SimpleRNN's total number of trainable weights must grow as the input sequence gets longer, since a longer sequence means more timesteps to process.' Is this claim correct, per how RNNs share parameters across time?

- A. The claim is correct only for LSTMs, but not for SimpleRNNs, which have a fixed weight count regardless of sequence length
 - B. Parameter count depends only on the number of training examples, not on sequence length or architecture at all
 - C. This claim is correct only when LOOKBACK exceeds 30 days, per the specific configuration used in Lab 08
 - D. No — an RNN applies the SAME weight matrices (for the input-to-hidden and hidden-to-hidden transformations) at EVERY timestep, regardless of sequence length; this parameter SHARING across time is precisely why RNNs can process sequences of varying and even previously-unseen lengths without changing the number of trainable weights
 - E. Yes — the claim is entirely correct; a distinct weight matrix is learned for each individual timestep in the sequence
-

85 A student differences the RAW cumulative series correctly, but forgets to first SUM the per-country/region rows for the US on days with multiple state-level rows, and instead accidentally differences each state row's cumulative count independently before any aggregation. What is the most likely consequence?

- A. The order of aggregation and differencing does not matter, since both operations commute perfectly regardless of the data's structure
- B. This mistake only affects the LSTM's performance, not the SimpleRNN's, since only the LSTM uses a cell state
- C. This mistake would cause the MinMaxScaler to fail with a runtime error, making the problem immediately obvious
- D. This mistake is equivalent to failing to clip negative values to zero, and requires the identical fix
- E. Differencing state-level rows independently before aggregating would produce a nonsensical daily series, since each 'difference' would be computed from an incomplete, inconsistent subset of the US total rather than from the correctly aggregated national cumulative count on each day — aggregation must happen BEFORE differencing, precisely because the raw file has multiple state rows per day that must first be summed

86 A GRU merges some of the functionality of an LSTM's separate forget and input gates into a single 'update gate.' What is the most accurate description of what the GRU's update gate does?

- A. It controls the balance between retaining the previous hidden state and incorporating new candidate information, functioning as a combined mechanism analogous to the LSTM's forget and input gates working together, rather than requiring two entirely separate gates and a separate cell state
- B. It has no relationship whatsoever to the LSTM's forget or input gates and serves a completely unrelated purpose
- C. It exclusively controls the OUTPUT of the GRU, exactly like the LSTM's output gate, with no role in updating the hidden state
- D. It doubles the number of parameters relative to an equivalent LSTM, since it must do the work of two gates at once
- E. It is identical in every respect to the LSTM's cell state itself, just under a different name

87 The glossary states MAE is 'more robust to outliers than RMSE.' A sudden one-day reporting anomaly causes a single extreme spike in the smoothed case-count series. How would this single anomalous day's error likely affect an RMSE-based evaluation differently from an MAE-based one?

- A. The anomalous day would only affect the naive baseline's error, not the LSTM's or SimpleRNN's error, since baselines are computed differently
- B. Because RMSE squares each error before averaging, a single very large error on the anomalous day contributes disproportionately to the overall RMSE (its influence grows quadratically with the error's size), whereas MAE's error contribution grows only linearly — so a single extreme outlier day inflates RMSE more than it inflates MAE
- C. RMSE and MAE would be affected identically by a single outlier day, since both metrics are computed as simple averages of the daily errors
- D. MAE would be MORE affected than RMSE by a single outlier day, the reverse of the standard robustness relationship
- E. Neither metric can be computed at all once a single outlier day is present in the series

88 Suppose a team correctly splits BY DATE (last 60 days as test) but constructs windowed training examples whose 30-day INPUT lookback is allowed to extend from the training period INTO the first ~30 days of the nominal test period, while only the TARGET day is required to fall in the training period. What subtle leakage issue, if any, does this introduce?

- A. This is leakage, but only because it affects the MinMaxScaler's fit, unrelated to the windowing itself
- B. This scenario is impossible to construct given how LOOKBACK=30 is defined, so the question describes a scenario that cannot occur
- C. This does introduce leakage: some training examples' target values would then be predicted using input windows that include days from what was meant to be an unseen, held-out test period, meaning the model has effectively already 'seen' those days' values during training — the entire windowed sequence (both lookback AND target) must respect the train/test date boundary
- D. There is no leakage issue at all, since only the target day's date matters for a valid split; the input window's date range is irrelevant to leakage
- E. This issue only matters for LSTMs, not SimpleRNNs, since only LSTMs have long-range memory

89 Vanishing gradients cause a SimpleRNN's early-timestep gradients to shrink toward zero over long sequences. What is the CONTRASTING failure mode called, in which gradients instead grow uncontrollably large during backpropagation through time, and what is a standard practical mitigation for it?

- A. Gradient saturation — the standard mitigation is to increase the learning rate to compensate for the excessively large gradients
- B. Gradient collapse — the standard mitigation is to remove the recurrent connections entirely, converting the RNN into a standard feedforward network
- C. Weight decay — the standard mitigation is to decrease the LOOKBACK window to exactly 1 day
- D. Exploding gradients — a standard mitigation is gradient clipping, which caps the gradient's magnitude (or norm) at a fixed threshold before the weight update is applied, preventing any single backward pass from producing a destructively large parameter update
- E. Also called vanishing gradients — the same term applies to both shrinking and growing gradients, since both represent gradient 'instability'

90 An LSTM maintains TWO separate recurrent quantities passed from one timestep to the next: the hidden state (h) and the cell state (c). Which statement correctly distinguishes their roles?

- A. The hidden state and cell state are two different names for exactly the same quantity, differing only by which variable name a particular code implementation happens to use
- B. The cell state is used only for the layer's final OUTPUT, while the hidden state is used only internally and never exposed outside the LSTM cell — the reverse of their actual roles
- C. SimpleRNN also maintains a separate cell state identical to the LSTM's, making this distinction irrelevant to comparing the two architectures
- D. The cell state is recomputed from scratch at every timestep with no dependence on its own previous value, unlike the hidden state
- E. The cell state (c) acts as the long-term memory 'conveyor belt,' modified only through the gated forget/input mechanisms and providing the additive gradient path central to learning long-range dependencies; the hidden state (h) is a filtered, gated 'view' of the cell state (via the output gate) used both as that timestep's output AND as part of the input to the gates at the NEXT timestep

Topic 9: Modern GenAI: Transformers, LLMs & Retrieval-Augmented Generation

Normalisation and embedding-space mismatch, chunk-overlap trade-off extremes, precise hallucination definition, retrieval-bottleneck diagnosis

91 Cosine similarity 'reduces to a dot product' when vectors are L2-normalised. If two embedding vectors are NOT normalised (arbitrary magnitudes), what specifically goes wrong if you use their raw dot product as a similarity/ranking score instead of true cosine similarity?

- A.** The raw dot product is influenced by both the vectors' ANGLE (their semantic alignment) and their MAGNITUDES, so a chunk with a large-magnitude embedding vector could rank as 'more similar' to a query purely because of its magnitude, even if a different, smaller-magnitude chunk is actually more semantically aligned (smaller angle) with the query — normalisation removes this magnitude confound
 - B.** Nothing goes wrong; the raw dot product and cosine similarity always produce identical rankings regardless of vector magnitudes
 - C.** The raw dot product cannot be computed at all for unnormalised vectors, and will raise a mathematical error
 - D.** Using the raw dot product would only affect the CHUNKING step, not the retrieval step, so ranking is unaffected
 - E.** This issue only arises for vectors with more than 1,000 dimensions, and is irrelevant to typical embedding sizes
-

92 A team uses 500-word chunks with a 100-word overlap (20% overlap). A colleague proposes increasing the overlap to 480 words (96% overlap) to 'maximise the chance no fact is ever split across a boundary.' What is the strongest objection to this proposal?

- A.** High overlap would only affect the LLM's generation step, not the retrieval step, so there is no real cost to maximising it
 - B.** Extremely high overlap causes most of the corpus's content to be duplicated across MANY nearly-identical chunks, which bloats the size of the index, wastes storage and retrieval compute, and can cause the same information to dominate the top-k retrieved results redundantly, crowding out genuinely distinct relevant chunks — overlap is meant to be 'modest,' trading off boundary-safety against exactly these costs
 - C.** There is no objection; overlap should always be maximised as close to 100% as possible for optimal retrieval quality
 - D.** Increasing overlap has no effect on index size, only on embedding computation time
 - E.** Overlap above 50% is mathematically undefined and cannot be implemented in a real chunking pipeline
-

93 A RAG system is evaluated with $\text{hit-rate@5} = 0.90$, $\text{faithfulness} = 0.85$, and $\text{END-USER SATISFACTION} = 0.40$ (separately surveyed). Given that both retrieval and generation appear to be performing reasonably well, what is the most defensible next step?

- A. The team should immediately lower the LLM's temperature toward 0, since temperature is the only lever that could explain low satisfaction
 - B. End-user satisfaction is not a valid metric for any AI system and should be disregarded entirely
 - C. Since both of the course's core RAG-specific diagnostics (hit-rate and faithfulness) look healthy, the low end-user satisfaction likely points to a DIFFERENT layer of the problem than retrieval or generation-faithfulness per se — e.g., answer relevance/usefulness, tone, latency, or UI/UX factors — so the investigation should broaden beyond the RAG pipeline's core retrieval/generation loop
 - D. The 0.40 satisfaction score proves hit-rate and faithfulness must have been computed incorrectly, since a low satisfaction score always implies low hit-rate and low faithfulness
 - E. The team should immediately rebuild the vector index, since any complaint about a RAG system is always a retrieval problem by definition
-

94 A team re-indexes their document corpus using a NEW, upgraded embedding model, but due to a caching bug, user QUESTIONS at retrieval time continue to be embedded with the OLD embedding model for several days. What is the most likely consequence, and why?

- A. This mismatch would only affect chunking, not retrieval, since chunking and embedding are entirely independent steps
 - B. This mismatch would improve retrieval quality, since combining two different embedding models provides an ensemble-like benefit
 - C. This mismatch would only be detectable by inspecting the LLM's final generated text, never by examining retrieval scores directly
 - D. Retrieval quality would likely degrade significantly, because cosine similarity is only a meaningful measure of semantic closeness when both vectors being compared live in the SAME embedding space; comparing a question vector from the OLD model against document vectors from the NEW model is comparing two different, generally incompatible coordinate systems, so similarity scores become largely meaningless
 - E. There is no consequence, since cosine similarity works identically regardless of which embedding model produced each of the two vectors being compared
-

95 Even with the LLM explicitly instructed to ‘use only the supplied context’ and temperature set to 0, a RAG system can still hallucinate. Which is the most direct explanation for how this can happen despite both safeguards?

- A. This scenario is impossible; temperature 0 combined with a context-only instruction is mathematically guaranteed to eliminate all hallucination
- B. This can only happen if the chunking step used a chunk size larger than 10,000 words, which never occurs in practice
- C. This can only happen if cosine similarity was replaced with Euclidean distance, which is unrelated to hallucination risk
- D. Hallucination is impossible once retrieval has occurred at all, regardless of whether the retrieved chunks are actually relevant
- E. Temperature 0 and a context-only instruction reduce but do not ELIMINATE hallucination risk: the retrieved chunks might not actually contain the answer (a retrieval failure) — in which case a low-temperature model instructed to answer from context may still produce a confident-sounding but unsupported answer rather than admitting the context is insufficient — or the model may simply fail to perfectly follow the instruction even when relevant context IS present

96 Fine-tuning and RAG are framed as alternative approaches to incorporating new knowledge. Under what circumstance, if any, would COMBINING both (a lightly fine-tuned model that ALSO uses RAG) make sense, rather than treating them as strictly mutually exclusive?

- A. Fine-tuning and RAG address different needs and can be complementary: fine-tuning can teach a model a desired STYLE, format, or specialised reasoning pattern (things that persist across many queries and are impractical to re-explain in every prompt), while RAG continues to supply up-to-date, citable FACTUAL content at query time — using fine-tuning to solve a knowledge-freshness problem alone would be the wrong tool, but that does not make fine-tuning useless for every purpose
- B. Combining the two approaches is never sensible under any circumstances, since the course frames them as strictly mutually exclusive with no valid combined use case
- C. Combining the two approaches is only possible if the fine-tuning data and the RAG corpus are byte-for-byte identical
- D. RAG must always be disabled whenever any fine-tuning has been applied to a model, since the two cannot coexist technically
- E. Fine-tuning and RAG solve the exact same problem in the exact same way, so combining them provides no additional benefit under any circumstances

97 Increasing k (the number of retrieved chunks supplied to the LLM) from 3 to 20 for every query. What is the most likely trade-off introduced by this change?

- A.** k has no relationship to prompt length, since the LLM only reads the top-1 result regardless of how many chunks are retrieved
 - B.** A larger k increases the chance that the genuinely relevant chunk IS somewhere among those retrieved (helping hit-rate), but also increases prompt length/cost, dilutes the proportion of the prompt that is actually relevant to the question, and can make it harder for the LLM to identify and rely on the truly relevant portion amid more irrelevant surrounding context — there is no free lunch in simply maximising k
 - C.** There is no trade-off; increasing k always strictly improves both retrieval and generation quality with no downside of any kind
 - D.** Increasing k only affects the embedding step, having no relationship to the LLM's prompt or generation step at all
 - E.** Increasing k always REDUCES hit-rate, since retrieving more chunks makes retrieval strictly harder
-

98 The glossary defines a hallucination as 'a fluent, confident, and factually wrong statement,' typically about something absent from the model's training data. Which of the following is technically NOT best described as a hallucination under this specific definition, even though it is still an undesirable model output?

- A.** A model confidently answers a question about a recent event that occurred after its training cutoff, getting the details wrong
 - B.** A model fabricates specific numeric details not present anywhere in the supplied context, stating them with high apparent confidence
 - C.** A model correctly states 'I don't have enough information in the provided context to answer that' when the retrieved chunks genuinely do not contain the answer — this is an accurate, appropriately hedged response, not a confident, fluent, WRONG statement, even though it may be unsatisfying to the user
 - D.** A model confidently states an incorrect statistic that appears nowhere in its training data or the retrieved context
 - E.** A model fluently asserts a fabricated citation to a paper that does not exist
-

99 A course's lecture-note PDF has a clear hierarchical structure (chapters, sections, sub-sections). A naive chunking strategy splits the document into fixed 500-word chunks WITHOUT regard for these structural boundaries, sometimes cutting a chunk mid-way through one sub-section and starting the next chunk mid-way through the following sub-section. What specific downstream problem does this structure-blind chunking risk, beyond the generic 'facts split across boundaries' issue already covered by overlap?

- A. This problem is identical to, and fully solved by, the same overlap mechanism that addresses facts split across boundaries
- B. This problem only affects the LLM's generation step, and has no relationship to the embedding or retrieval steps
- C. This problem can only be caused by chunk sizes larger than 2,000 words, and never occurs with 500-word chunks
- D. A single chunk may end up containing the TAIL of one topic and the HEAD of an entirely unrelated topic concatenated together, producing an embedding that represents a blend of two unrelated ideas rather than a coherent single topic — this is distinct from a single fact being split (which overlap helps with), since here the issue is topical incoherence WITHIN one chunk, which overlap alone does not fix
- E. This is not actually a problem at all; document structure is entirely irrelevant to chunking quality in a well-designed RAG system

100 Lab 09's RAG system is implemented with plain numpy rather than a dedicated vector database, and retrieval over the whole index is achievable via a SINGLE matrix multiplication (when vectors are unit-normalised). What is the practical SCALING limitation of this approach that a dedicated vector database is specifically designed to address?

- A. There is no scaling limitation whatsoever; a numpy matrix multiplication scales identically regardless of corpus size, from ten chunks to ten billion chunks
- B. The numpy approach is actually always faster than any dedicated vector database, regardless of corpus size, making vector databases entirely unnecessary in all cases
- C. The limitation only concerns the CHUNKING step, not the similarity search step, and is unrelated to corpus size
- D. The numpy approach cannot handle L2-normalised vectors at all, which is why cosine similarity cannot be computed this way
- E. A single matrix multiplication computing similarity against every stored chunk is an EXACT, brute-force search whose cost grows linearly with the number of chunks in the index; for very large corpora, this becomes computationally expensive per query — exactly the scaling problem that approximate nearest-neighbour structures in dedicated vector databases are built to solve, trading a small amount of exactness for much faster lookup at scale

101 A team's RAG system for querying internal fintech policy documents shows: $\text{hit-rate@5} = 0.55$ (comparatively low), $\text{faithfulness} = 0.92$ (comparatively high, when the right chunk IS retrieved). A stakeholder proposes: 'Since faithfulness is so high, let's focus all our effort on prompt engineering to push faithfulness even closer to 1.0.' Evaluate this proposal.

- A.** This proposal misallocates effort: with hit-rate@5 already the clear bottleneck (0.55, meaning the correct chunk is missing in nearly half of queries) and faithfulness already strong when retrieval does succeed, further prompt engineering can only ever improve outcomes for the $\sim 55\%$ of queries where retrieval already succeeds — it cannot fix the $\sim 45\%$ failing due to retrieval, so the team should prioritise retrieval improvements instead
 - B.** The proposal is sound; faithfulness should always be prioritised over hit-rate regardless of their relative values, since faithfulness is the more important of the two metrics by definition
 - C.** The two metrics are actually measuring the same underlying phenomenon, so improving either one through any means will proportionally improve the other
 - D.** A hit-rate@5 of 0.55 is already excellent and requires no further attention, since any hit-rate above 0.50 is considered a passing threshold
 - E.** Prompt engineering is capable of directly improving hit-rate@5 as effectively as it improves faithfulness, so the proposal is equally valid either way
-

Answer Key

Correct answers and brief rationale for every question.

Topic 1: Foundations & End-to-End ML Pipelines

- 1. A** The syllabus handout (Section 6, Academic Assessment) is the real course record; the RAG corpus is explicitly built as practice material for Lab 09's retrieval exercise. Nothing indicates it supersedes the syllabus — this tests whether the reader conflates a lab's synthetic knowledge base with the actual course record.
- 2. B** Overfitting's signature is exactly this: low training error with a substantial gap to a higher test error. Option B describes underfitting (high bias) instead — a common confusion this question is designed to catch.
- 3. C** Options B-E are the course's own three practical habits the theorem licenses; option A is precisely the nihilistic misreading the course warns against — treating the theorem as license for arbitrary model choice rather than for consciously chosen, empirically validated priors.
- 4. D** This is textbook CACE/entanglement: because weights are shared across features via learned interactions, a change to one feature's representation can silently redistribute the model's reliance across every other feature, even when nothing else was 'touched.'
- 5. E** This is the hidden-feedback-loop trap in its sharpest form: it isn't merely 'a minor limitation' (option B) — it's that no ground truth exists at all for declined applicants, so no honest false-negative rate can be computed for that group in the first place.
- 6. A** Configuration drift silently changing a feature-store window is a realistic, well-documented failure category from the 'small green box' framing; the other options implausibly swap cause and effect between unrelated system layers.
- 7. B** The table's exact framing: batch poisoning 'must survive a retrain review'; a poisoned stream in online learning 'corrupts the live model' directly — option B reverses this.
- 8. C** Data must be collected/prepared before candidate families can be trained and shortlisted, and hyperparameter fine-tuning and final selection follow that shortlisting — the natural, and course-specified, order.
- 9. D** $C = 1/\lambda$, so larger C means weaker regularization (smaller λ); this trades lower training loss for a bias-variance trade-off whose net effect on test risk is not guaranteed to be positive or negative — it depends on where the model already sits.
- 10. E** This is a 'none of the above' trap: every one of these listed items is drawn as a system component SURROUNDING the small green ML-code box, not as part of the core itself — recalling just one as 'part of the core' would be a reading error.
- 11. A** The table's exact pairing: batch — credit scoring, limit setting; online — streaming fraud on payment rails, matching the theme of validation/audit strength (batch) versus fast-drift tracking (online).
- 12. B** This scenario is CACE at the level of individual-row rankings rather than an aggregate metric: an aggregate score (AUC) can stay flat while the model's internal reliance on correlated features shifts enough to change fairness properties — the essence of entanglement.

Topic 2: Advanced Data Preparation & Feature Engineering

13. C MAR means missingness is explainable by OTHER observed variables. Option B actually describes MNAR (missingness tied to the variable's own unobserved value) — a classic confusable trap between the two mechanisms.

14. D Leakage is formally a content mismatch — ϕ_{train} has access to information ϕ_{serve} genuinely cannot have at prediction time — not a superficial difference like language, column count, or ordering.

15. E A larger prior strength m increases the shrinkage weight toward the global mean, which matters most for rare/low-sample categories whose own raw statistic is noisiest.

16. A Resolving entities correctly does not, by itself, force customer-level grouping during cross-validation — that requires GroupKFold specifically; without it, one customer's rows can straddle the train/validation boundary.

17. B The period parameter must match the variable's genuine cyclical length; using the wrong period (e.g. reusing 24 for a 7-value cycle) would misrepresent the adjacency structure entirely.

18. C This is the subtler 'future leaks into past' violation, distinct from the textbook scaler example — a feature computed over a customer's ENTIRE history (including future transactions relative to a given row's scoring time) violates temporal leakage-safety even though no train/test boundary is crossed in the classic sense.

19. D WoE is defined from the log-ratio of good-to-bad odds relative to the population; a large positive value signals a disproportionately 'good' (low-risk) bin, unrelated to row count, missingness, or the raw value's sign.

20. E The offset trick introduces a hidden, data-dependent parameter (the offset itself) that must be tracked consistently between train and serve — exactly the kind of instability Yeo-Johnson avoids by handling zero/negative values directly.

21. A Statistical improvement is irrelevant if the improvement comes from information that would not have existed at scoring time — this is temporal leakage regardless of which feature it touches.

22. B Deduplication removes redundant identical records; entity resolution groups distinct, legitimate records under a shared true identity — different operations addressing different data quality issues.

23. C This weighs the two genuine, course-specific concerns against each other: dimensionality explosion (favouring target encoding) versus the self-leakage risk of naive target encoding (requiring the out-of-fold fix).

24. D Only the manual_review_score/settlement_status leak has an explicitly quantified magnitude ($\sim +0.37$ ROC-AUC) in the course material — the largest single documented effect among the three candidate issues.

Topic 3: Classical ML Evaluation & Imbalanced Classes

25. E The no-skill baseline for PR-AUC tracks prevalence, not 0.5; comparing 0.35 to an absolute '1.0 is perfect' scale, without accounting for a ~ 0.06 baseline, badly understates the model's actual lift over chance.

26. A Threshold optimisation against a cost matrix is jointly a function of costs AND the underlying score distribution/class balance; the specific ~ 0.23 figure is tied to this dataset's $\sim 6\%$ rate,

not a universal constant of the 10:1 ratio alone.

27. B Discrimination (ROC-AUC) and calibration are orthogonal properties; a high-ROC-AUC, overconfident model is exactly the situation CalibratedClassifierCV is designed to correct, without needing to retrain the underlying model.

28. C Each scheme guarantees only its own property (stratification OR grouping), not both simultaneously — a dataset needing both requires a combined stratified-group splitting strategy.

29. D $F1(A) = 2(0.9)(0.1)/1.0 = 0.18$; $F1(B) = 2(0.5)(0.5)/1.0 = 0.50$. Both have arithmetic mean 0.50, but F1's harmonic-mean structure heavily penalises Model A's large precision-recall imbalance.

30. E ROC-AUC is rank-based; any strictly monotonic transform of the scores leaves every pairwise ranking — and therefore ROC-AUC — exactly unchanged.

31. A None of these standard metrics natively encodes an asymmetric business cost; the lab's own methodology is exactly to sweep the threshold against an explicit cost matrix rather than trust any single symmetric metric to capture it.

32. B $10,000 \text{ rows} / 5 \text{ folds} = 2,000 \text{ rows per fold}$; $6\% \text{ of } 2,000 \approx 120 \text{ positive examples per fold}$ under stratification.

33. C Below-diagonal at high predicted probability means predicted > observed, i.e. overconfidence — the mirror-image pattern from a curve above the diagonal (underconfidence); the two are not interchangeable.

34. D The threshold is downstream of the cost matrix, not of the model's learned weights; a change in real-world costs calls for re-sweeping the threshold against the updated cost matrix, not retraining the model itself.

35. E The rebuttal must invoke the different no-skill baselines of the two metrics; taken against the correct ~ 0.06 baseline for this dataset, 0.30 signals meaningful lift that the naive 0.63-vs-0.30 framing hides.

Topic 4: Ensemble Methods & Hyperparameter Tuning

36. A Bootstrap samples overlap (drawn from the same underlying training set), inducing positive correlation between trees; the σ^2/B formula assumes full independence and therefore understates the true variance of the ensemble average.

37. B $(1 - 1/n)^n$ converges to $1/e \approx 0.368$ as $n \rightarrow \infty$, and this convergence is already very close to the limit for moderately sized n — this is why $\sim 36.8\%$ is a stable rule of thumb across dataset sizes.

38. C AdaBoost's exponential loss aggressively up-weights confidently-wrong examples at every stage; a persistently mislabeled row keeps receiving exponentially growing weight, distorting subsequent weak learners around it.

39. D AdaBoost predates the gradient/residual view of boosting but expresses the same sequential-correction idea via row reweighting — historically first, conceptually related, not a different family (bagging) or a later refinement.

40. E Row bootstrapping alone does not stop every tree from repeatedly choosing the same dominant feature at the top split; feature subsampling specifically forces trees to diversify their reliance on features, an effect row sampling alone cannot replicate.

- 41. A** TPE's defining feature is exactly what Student A describes: probabilistic modelling of good vs. bad configurations from past trials, guiding where to sample next — this is what makes it more sample-efficient than exhaustive or purely random search.
- 42. B** Pruning saves compute by cutting off unpromising trials mid-training rather than fully completing them, reallocating that saved budget toward exploring more configurations within the same total time/compute budget.
- 43. C** This is the documented Gini-importance bias toward high-cardinality features; permutation importance, computed on held-out data by directly measuring performance impact, does not share this bias — making it the more trustworthy of the two in such a disagreement.
- 44. D** `subsample/colsample_bytree` introduce stochastic row/column sampling per boosting round; `reg_alpha/reg_lambda` are direct L1/L2 penalty terms in the objective — two functionally distinct regularisation mechanisms.
- 45. E** OOB randomises rows, not time; a one-year-later test set can reveal real drift that a row-wise OOB estimate, computed entirely within the training period, is structurally incapable of detecting.
- 46. A** Bagging's mechanism (averaging) addresses variance, not bias; shallow (high-bias) trees leave a bias floor that no amount of averaging removes — deep, low-bias trees are the natural fit for bagging's variance-reduction strength.

Topic 5: Unsupervised Learning & Dimensionality Reduction

- 47. B** Inertia is monotonic non-increasing in k by construction (more clusters can only reduce or match within-cluster distance); the elbow method therefore looks for a bend in the rate of decrease, not a minimum, since there isn't one to find before k equals the sample size.
- 48. C** Elbow and silhouette measure genuinely different properties; a legitimate disagreement calls for substantive investigation of both candidate solutions, not blind priority of one diagnostic.
- 49. D** A too-small ϵ starves most points of enough neighbours to qualify as core points, flooding the result with noise labels — the opposite direction from a too-large ϵ , which instead over-merges clusters.
- 50. E** PCA is unsupervised and transforms all features into new combinations, ignoring the target; target-correlation selection is supervised and simply keeps/drops original features — categorically different operations.
- 51. A** The explained variance ratio for the top d components is the sum of their eigenvalues over the trace (total variance): $(8+4)/15 \approx 0.80$, i.e. 80%.
- 52. B** t-SNE's sensitivity to hyperparameters like perplexity is a known, expected property, reinforcing why its output is treated as visualisation-only rather than a stable feature representation, unlike PCA's linear, deterministic projection.
- 53. C** High reconstruction error signals that a point is poorly represented by the retained low-dimensional structure — a candidate signal for further investigation, not a definitive verdict about the point's validity or label.
- 54. D** Hierarchical clustering's dendrogram is computed once and can be cut at any height afterward to yield any number of clusters, unlike K-Means, which must be rerun from scratch for each candidate k .
- 55. E** A negative silhouette score specifically means the average distance to the nearest OTHER

cluster is smaller than the average distance to the point's OWN cluster — a mis-assignment signal, not an outlier flag per se, and unrelated to DBSCAN's separate noise-labelling mechanism.

56. A Raising `min_samples` raises the bar for a point to qualify as a core point at a given `eps`, generally shrinking and thinning clusters and pushing more borderline points into the noise label.

57. B This combines two separate concerns: a 2-component PCA used purely for visualisation may capture far less than the 95%-variance threshold used elsewhere, undermining the visual impression; and even a genuine cluster structure would still require supervised validation, not deployment on visual inspection alone.

Topic 6: Deep Learning Fundamentals (MLP)

58. C Standard (inverted) dropout is disabled at inference, with outputs rescaled during training to keep expected activation magnitudes consistent — inference uses the full deterministic network, not a randomly thinned one.

59. D Because the chain rule multiplies per-layer derivatives, even the best-case sigmoid derivative of 0.25 compounds geometrically across depth ($\approx 0.25^L$), driving gradients toward zero in deep networks — the mechanistic root of vanishing gradients.

60. E `m` (first moment) smooths the gradient direction like momentum; `v` (second moment) rescales the effective step size per parameter, dampening updates for parameters with historically large squared gradients — this is Adam's actual, and correctly stated, design.

61. A A single sigmoid output structurally cannot represent a 3-way categorical distribution; K mutually exclusive classes require K softmax units with categorical cross-entropy, exactly the distinction the course draws between binary and multiclass setups.

62. B A patience of 1 is overly sensitive to normal epoch-to-epoch validation-loss noise, risking premature stopping before the true (still-improving) trend has played out — exactly why a more tolerant patience like 10 is used.

63. C `class_weight` is a loss-reweighting mechanism operating on the existing dataset; SMOTE instead alters the training set's composition by generating synthetic minority examples — genuinely different mechanisms for the same underlying goal.

64. D The course's statement is a specific empirical observation on one tabular dataset, not a universal claim about all datasets, data types, or hardware — over-generalising a single documented comparison into a universal law is the flaw.

65. E L2 penalises weight magnitude directly in the loss; dropout instead randomly disables units to prevent co-adaptation — mechanistically distinct approaches, which is exactly why the lab combines them rather than treating either as redundant.

66. A ReLU's unbounded $[0, \infty)$ range cannot represent a probability; sigmoid's $[0, 1]$ bound is required for `binary_crossentropy` — this is exactly why ReLU is right for hidden layers but wrong for this output layer.

67. B The same fit-on-train-only discipline applies here as everywhere else in the course: the scaler's parameters are learned once from training data and applied unchanged to every subsequent dataset, never refit on validation, test, or production data.

68. C Matched, high, early-plateauing loss is the underfitting signature, not overfitting; the regularisation toolkit (dropout/L2/early stopping) is designed to address the OPPOSITE failure mode and offers no mechanism to explain or fix this pattern.

Topic 7: Convolutional Neural Networks & Computer Vision

69. D $\text{floor}((64+0-5)/2)+1 = \text{floor}(59/2)+1 = 29+1 = 30$ — careful application of the standard output-size formula, term by term.

70. E Receptive field grows by $(\text{kernel_size} - 1)$ per stacked stride-1 layer: $3 + (3-1) = 5$, not the naive sum (6) or product (9) of the kernel sizes.

71. A Each filter has $3 \times 3 \times 3 = 27$ weights plus 1 bias = 28 parameters; with 32 filters, total = $28 \times 32 = 896$.

72. B `MaxPooling2D(2x2)` halves each spatial dimension while keeping the channel count ($14 \rightarrow 7$, so $7 \times 7 \times 256$); `GlobalAveragePooling2D` collapses the entire spatial extent per channel down to one value each, yielding a flat 256-length vector.

73. C 'Frozen' specifically means the optimiser skips WEIGHT updates for those layers; gradient computation/flow through them can still occur as needed for the computation graph, a subtlety often glossed over in casual descriptions of freezing.

74. D The key distinction is categorical, not just a matter of degree: rotation preserves left-right anatomy while modelling realistic positioning variance, whereas a flip structurally violates left-right anatomy regardless of magnitude.

75. E Each 'valid' 3×3 stride-1 convolution shrinks spatial size by $(f-1)=2$: $32 \rightarrow 30 \rightarrow 28 \rightarrow 26$ across the three layers, illustrating cumulative shrinkage — a real design concern in deep 'valid'-padded stacks.

76. A The direction follows directly from what has something to lose: pretrained weights carry useful, transferable knowledge that a large update could destroy, while randomly-initialised head weights have nothing yet to protect.

77. B The confusion matrix arithmetic given is actually correct ($\text{accuracy} = 180/200 = 90\%$); the critique is about the metric's blindness to the clinically-asymmetric cost of the 15 missed pneumonia cases specifically.

78. C Augmentation multiplies the effective variety seen from the SAME underlying small sample, reducing exact-memorisation risk — it does not create new unique patients, and its benefit is about generalisation, not raw dataset expansion or speed.

79. D 'Same' padding keeps each `Conv2D` layer spatial-size-neutral; only the five stride-2 pooling layers reduce resolution: $256 \rightarrow 128 \rightarrow 64 \rightarrow 32 \rightarrow 16 \rightarrow 8$, a cumulative downsampling (total stride) of $2^5=32$.

Topic 8: Sequence Models: RNNs, LSTMs & Time-Series Forecasting

80. E With a sliding window of length 30, the number of valid (X,y) pairs from a 400-day series is $400 - 30 = 370$, since the window needs 30 prior days available before it can predict the next one.

81. A A window is entirely within the 60-day test period only once both its 30-day lookback and its target day sit inside that window — the first 30 test-period days cannot supply a fully-inside lookback, leaving $60 - 30 = 30$ valid entirely-in-test windows.

82. B A rolling average mechanically increases day-to-day similarity by construction, which is exactly why 'predict yesterday's value' becomes a stronger baseline on the smoothed series than it would be on the noisier raw series.

- 83. C** The forget gate governs retention/discarding of the EXISTING cell state; the input gate governs how much NEW candidate information is written in — complementary, not identical or reversed, roles.
- 84. D** Weight sharing across timesteps is the defining structural property of an RNN — the exact same weight matrices are reused at every step, which is why the parameter count is independent of sequence length.
- 85. E** Aggregation and differencing do NOT commute here: differencing scattered state-level rows before summing them into a national total produces meaningless values, since the correct national daily change requires summing first, then differencing.
- 86. A** The GRU's update gate is precisely a merged retain-vs-incorporate mechanism, doing the combined work of the LSTM's separate forget and input gates with fewer parameters, not more — consistent with the GRU's description as a 'lighter variant.'
- 87. B** RMSE's squaring of errors makes it grow quadratically with a single large error, disproportionately inflating it; MAE's linear error contribution is comparatively robust — the mechanistic explanation behind the glossary's stated fact.
- 88. C** The whole windowed example (lookback AND target) must respect the date boundary, not just the target day — allowing the lookback to reach into the nominal test period leaks future information into training regardless of where the target day itself falls.
- 89. D** The contrasting failure mode is exploding gradients; gradient clipping (capping the gradient's magnitude/norm) is the standard practical mitigation, distinct from any of the other proposed remedies.
- 90. E** The cell state is the gated long-term memory pathway (the additive gradient path); the hidden state is the output-gate-filtered view of it used both as output and as the recurrent input to the next step's gates — a distinction SimpleRNN has no analogue for, since it lacks a separate cell state entirely.

Topic 9: Modern GenAI: Transformers, LLMs & Retrieval-Augmented Generation

- 91. A** Without normalisation, the dot product conflates magnitude with directional (semantic) alignment, so a purely larger-magnitude chunk vector could outrank a more semantically relevant, smaller-magnitude one — exactly the confound normalisation removes.
- 92. B** The course frames overlap as 'modest' precisely because near-total overlap trades away index efficiency and retrieval diversity for a marginal boundary-safety gain — maximising it has real, named costs, not a free improvement.
- 93. C** With both RAG-specific diagnostics healthy, the diagnostic principle points AWAY from retrieval/generation as the culprit and toward other layers (relevance, tone, latency, UX) that hit-rate and faithfulness simply were not designed to capture.
- 94. D** Comparing vectors from different embedding spaces is comparing incompatible coordinate systems; similarity scores computed this way are largely meaningless, directly degrading retrieval quality — this is exactly why the query and document embedder must be 'the same embedding model.'
- 95. E** Neither safeguard is an absolute guarantee: a retrieval failure (irrelevant chunks) can still be answered confidently despite low temperature, and instruction-following is itself imperfect — hallucination risk is reduced, not eliminated, by either mechanism.

- 96. A** The course's critique of fine-tuning targets its unsuitability for the KNOWLEDGE-FRESHNESS problem specifically — it does not claim fine-tuning is useless for everything; style/format/reasoning-pattern training and RAG's factual grounding address genuinely different needs and can coexist.
- 97. B** Retrieving more chunks can only help or maintain hit-rate (never hurt it, by definition of a superset), but it comes at the real cost of prompt bloat and dilution of relevant signal — a genuine, correctly-directed trade-off, not a free improvement.
- 98. C** By the definition's own terms (confident AND wrong), an accurate 'I don't know' response is neither confident-and-wrong nor a fabrication — it is the OPPOSITE of a hallucination, even though it may leave the user unsatisfied; the other four options all match the definition's criteria.
- 99. D** Overlap fixes facts SPLIT at a boundary; it does nothing for a chunk that MERGES two unrelated topics together into one incoherent embedding — a genuinely distinct problem requiring structure-aware chunking, not more overlap.
- 100. E** Brute-force matrix-multiplication search is exact but scales linearly with index size; approximate nearest-neighbour structures in vector databases exist precisely to trade a bit of exactness for much better scaling at large corpus sizes.
- 101. A** The bottleneck logic is decisive here: prompt engineering operates entirely downstream of retrieval and can only help the ~55% of queries retrieval already serves correctly — it structurally cannot fix the ~45% failing at the retrieval stage, so effort belongs there instead.